The inner membrane of mitochondria is impermeable to ions, particularly to protons, which accumulate outside the membrane, thereby creating an electrochemical potential difference across the membrane ($\Delta\mu H^+$). This is called as chemical potential and electric potential.

The chemiosmotic coupling theory has the following features:

1. As electrons pass through the ETC, protons are transported from the matrix and released into the intermembrane space. An electrical potential (ψ) and proton gradient (ΔpH) arise across the inner membrane. The electrochemical proton gradient is called as proton motive force.
2. Protons, now in greater excess in the intermembrane space, pass directly through the inner membrane and back to the matrix, down the concentration gradient, via specific channels, thereby producing ATP.

8.3.2 Role of ATP synthase

The enzyme complex ATP synthase is known as Complex V. It synthesizes ATP using the energy of the proton gradient generated by the electron transport chain. The role of this ATPase is to bring back the protons to the mitochondria. The interior of a mitochondrion is alkaline and the reaction is favourable to drive the synthesis of ATP.

The proton-flow through the F_1F_0 ATPase is required to release ATP from the active site where it was synthesized from ADP and P_i. The ATP made in the interior of the mitochondria must be exchanged for ADP outside the mitochondria to keep the cytosol supplied with ATP. The exchange of mitochondrial ATP for cytoplasmic ADP is catalyzed by the ATP/ADP translocase. The complete transfer of 2 electrons from NADH through the entire ETC to oxygen generates 3 ATPs. One $FADH_2$ feeds electrons into coenzyme Q after the first ATP-generating step. Flavin-linked substrates generate only 2 ATPs per 2 electrons transferred down the chain. Flavin-linked substrates generate only 2ATPs, not only because they feed electrons after the first ATP has already been made, also because $FADH_2$ is not as strong a reducing agent as NADH. There is not enough energy in the oxidation of $FADH_2$ to generate 3 ATPs.

8.4 HIGH ENERGY COMPOUNDS

High energy compounds are usually referred to as high-energy phosphates. Inorganic phosphate groups form high energy bonds which have the ability to break, provide energy, and run the metabolic processes of life. The hydrolysis of high energy phosphate bonds releases energy. For a reaction to be feasible, ΔG (Change in Gibbs free energy) value must be negative. Compounds that contain phosphate are known as high-energy entities with a large negative value of ΔG of the order of -25 or -30 kJ/mol.

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